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Layer-wise Representation Dynamics: An Empirical Investigation Across Embedders and Base LLMs

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Abstract

Hidden states change substantially across the layers of modern language models, but most layer-wise analyses focus on one aspect of that change. We propose Layer-wise Representation Dynamics (LRD), a framework with three layer-wise measurement families: Frenet (Grassmann speed and curvature) for global subspace motion, Neighborhood Retention Score (NRS) for local nearest-neighbor retention, and Graph Filtration Mutual Information (GFMI) for alignment with the final layer. Applying LRD to 31 models (encoder-based and decoder-based embedders, plus base LLMs) on 30 MTEB tasks reveals architectural and task-level differences that are not apparent from final-layer representations alone. We then use LRD for two applications: label-free model selection and inference-time layer pruning. For selection, all three model-level scores correlate positively with downstream MTEB performance, with end-to-end subspace displacement ($d_{0,L}$) the strongest, and the same direction holds on a smaller base-LLM MMLU panel. For pruning, GFMI is the only measurement-guided rule that beats Random at the 15% and 20% budgets and has the best median change at every budget. Frenet is effective only at the lightest budget, while NRS does not transfer from model selection to pruning. These results show that layer-wise structure provides signal for both interpretation and deployment decisions.

1 Introduction

Modern language models do not form their hidden representations in the same way. Embedders (encoder-based or decoder-based) and base large language models (LLMs) may organize the same examples through different layer-wise structures before producing their final-layer states. Understanding these structures matters because the semantic space is shaped across layers, not only by the final embedding. Representation-similarity methods make hidden states comparable across layers and models [31, 73], while recent layer-wise analysis shows that intermediate layers can reveal structure that is not captured by the final layer alone [60]. We therefore analyze the layer-wise representation sequence each model produces on the same inputs, and compare these sequences across models from multiple perspectives.

Comparing these sequences from multiple perspectives in a unified way is still uncommon. Layer-wise studies have made the sequence visible but typically emphasize one criterion, such as a probe, a similarity score, or a grouping rule [29, 18, 14, 39]. Methods that use representations as evidence, such as label-free model selection [11], usually rely on a single representation, commonly the final embedding, and rarely incorporate the layer-wise view. The formation process is therefore underused both for interpretation and for representation-based decisions.

Motivated by these observations, we introduce Layer-wise Representation Dynamics (LRD), an analysis built from three complementary layer-wise measurement families. Each family targets a

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distinct aspect of the layer sequence, and the three are complementary rather than redundant. One family measures global subspace motion, another local nearest-neighbor retention, and the third alignment with the final layer. For each model-task pair, LRD follows the same examples through the network, making it possible to distinguish layers that redirect the representation globally, maintain local neighborhoods, or already resemble the final-layer organization.

We apply LRD to 31 off-the-shelf models on 30 MTEB tasks. The pool contains 25 embedders (encoder-based and decoder-based) and 6 base LLMs. The tasks include classification, retrieval, semantic textual similarity, reranking, and duplicate detection. Across this set of model-task pairs, we can compare layer-wise behavior between architectures and across task types.

Beyond analysis, we use LRD for model selection and layer pruning. Our contributions are threefold. (1) We introduce LRD, a framework of three complementary measurement families that capture global subspace motion, local-neighborhood retention, and final-layer alignment across layers. (2) LRD reveals architectural and task-level differences that final-layer representations alone do not show. (3) The LRD measurements inform two applications: model selection on MTEB, where the three model-level scores correlate positively with downstream performance and $d_{0,L}$ is strongest, and layer pruning, where GFMI beats Random at the 15% and 20% budgets and has the best median change at every budget.

Section 2 reviews related work on layer-wise analysis, representation-based assessment, and the foundations of LRD. Section 3 defines the three measurement families. Section 4 describes the model-task setup and family-level observations, Sections 5–6 present the two application studies, and Section 7 discusses limitations and concludes the paper.

2 Related Work

2.1 Layer-wise representation analysis

Intermediate layers of modern language models carry linguistic and semantic structure rather than acting as uniform stages between input and output. Probing studies on BERT and related encoders locate syntactic and semantic information at specific depths [65, 25]. Best-layer and progression analyses add that semantic signals are not always strongest at the final layer and that hidden-state sequences fall into recognizable layer groups [60, 29, 18]. A complementary line of work documents concrete layer-wise phenomena: objective-dependent bottom-up accumulation [69], anisotropy and narrow-cone behavior at deeper layers [13, 17], and broader transformer-geometry observations on subspace structure, phase-like changes, and trajectory shape [55, 68, 27, 37]. Together these results establish the layer sequence as an informative object of study and motivate the layer-wise analysis we develop with LRD.

Across this body of work, however, the chosen unit of analysis differs substantially: a probe target, a similarity score between two layer matrices, a coarse layer block, or a geometric quantity such as subspace orientation or curvature. Each choice answers its own question well, but their evidence has rarely been brought together across multiple model families and task types under one protocol. LRD does not propose yet another single criterion. It specifies a uniform layer-wise extraction protocol and reports three complementary measurement families on the same model-task pairs, so that cross-model and cross-task comparisons rest on a common measurement basis.

2.2 Representation-based model assessment

Internal representations also serve as evidence for model assessment before full task-specific evaluation. Representational similarity analysis (RSA) and its neural-network extensions compare systems through pairwise relations among examples or through learned representation matrices [31, 73, 32, 54, 47, 10]. Alignment-based criteria ask whether an embedding space is consistent with an external linguistic or structural reference [67, 22]. Effective-rank methods such as RankMe use the spectrum of learned representations as an unsupervised proxy for downstream performance [19], and information-sufficiency criteria such as EMIR rank embedders by representation-level signals before benchmark evaluation [11]. These methods establish that representation structure can carry informative pre-evaluation signal, but the evidence is typically a single representation per model, most often the final embedding. The formation process by which that embedding is reached is largely set

aside. LRD's model-selection study uses this formation process directly: it treats the full layer-wise sequence on the same examples as label-free evidence for downstream MTEB performance.

2.3 Metric foundations and roles

LRD combines three measurement families as complementary diagnostics under a shared layer-wise analysis protocol. We do not claim a single underlying theory that unifies them into one score. Each family targets a different kind of layer-wise change and is grounded in its own line of prior work. The preceding sections motivate two constraints on LRD: representations should be followed across depth, and the resulting evidence should not be collapsed into a single holistic similarity value. Additional layer-wise probing work reinforces this point by showing that the informative layer can depend on feature type, architecture, and downstream use [40, 28, 15, 78]. For LRD, the implication is not to choose another best layer or another single similarity score, but to keep distinct measurement roles. Dominant-subspace rotation, local-neighborhood reordering, and intermediate-to-final alignment are different events, even when they produce similar aggregate similarity. To isolate the global subspace component, LRD adopts Grassmannian geometry, where principal-angle distances give a standard notion of subspace displacement [1]. This choice is consistent with prior transformer-geometry analyses of how hidden-state structure changes across layers [55, 68]. The Grassmann speed and curvature in LRD belong to this subspace role and are not intended as proxies for local or graph-level structure.

For the local component, manifold-learning and graph-regularization methods represent local structure through neighborhood graphs [64, 3, 4], a view directly relevant to embedding-based retrieval and matching, where neighborhood structure and embedding-space distances matter for task performance [48, 56]. The Neighborhood Retention Score (NRS) builds on this lineage. For the final-layer alignment component, Graph Filtration Mutual Information (GFMI) combines three traditions: alignment-style comparison [67, 22], filtration-based multi-scale graph comparison [12], and mutual-information partition comparison [52]. The goal of bundling these three families is not to collapse them into one number, but to retain their distinct sensitivities within a single evaluation protocol.

3 Layer-wise Measurement Families

In this section, we define three measurement families that quantify how representations evolve across layers. For each task, we fix a set of N examples, indexed by $i = 1, \ldots, N$. For a model with layers $0, \ldots, L$, layer l produces a representation matrix $X_l \in \mathbb{R}^{N \times d}$, where d is the representation dimension and the i-th row of X_l is the representation of example i at layer l. The measurements are computed on the discrete matrix sequence $X_0, X_1, \ldots, X_L$.

3.1 Global subspace motion: Grassmann speed and curvature

The Frenet measurement family represents each layer by its dominant linear subspace and treats the layer sequence as a discrete path in the space of subspaces, with adjacent-layer distance and three-layer curvature playing the roles of speed and bending. We draw on the subspace reduction used by Raghu et al. [54] and compare centered representation matrices through their leading singular subspaces. For each model-task pair, we fix the subspace dimension r as the smallest rank that explains 95% of the variance of the final-layer representation X_L. We use the final layer because X_L is the representation used by downstream tasks, and we apply the same r to every layer so that all subspaces lie in $\mathrm{Gr}(d, r)$, the Grassmann manifold of r-dimensional subspaces of $\mathbb{R}^d$. For layer l, $Q_l \in \mathbb{R}^{d \times r}$ denotes an orthonormal basis for this subspace, given by the top-r right singular vectors of the centered X_l.

Distances between layers are then computed on $\mathrm{Gr}(d, r)$. If $\theta_1, \ldots, \theta_r$ are the principal angles between Q_a and Q_b, we use the Grassmann geodesic distance [1]

$$d_{\mathrm{Gr}}(Q_a, Q_b) = \left(\sum_{i=1}^{r} \theta_i^2 \right)^{1/2}, \qquad \theta_i = \arccos \sigma_i(Q_a^\top Q_b). \tag{1}$$

Here $\sigma_i(\cdot)$ denotes the i-th singular value. Write $d_{a,b} = d_{\mathrm{Gr}}(Q_a, Q_b)$ for the distance between layers a and b. The adjacent-layer speed is $s_l = d_{l,l+1}$. Large s_l means that the dominant directions still change from layer l to layer $l + 1$. Small s_l indicates that the global subspace is relatively stable.

To capture whether the subspace path changes direction, we also compute Menger curvature over triples of consecutive layers. The curvature assigned to the middle layer is

$$\kappa_{l+1} = \frac{4A}{d_{l,l+1} d_{l+1,l+2} d_{l,l+2}}, \tag{2}$$

where A is the triangle area computed from sides $a = d_{l,l+1}$, $b = d_{l+1,l+2}$, $c = d_{l,l+2}$ via Heron's formula, $A = \sqrt{p(p-a)(p-b)(p-c)}$ with $p = (a+b+c)/2$. Degenerate triples, such as zero-area triangles, are assigned zero curvature, matching the implementation. The next measurement family shifts from global subspaces to local neighborhoods.

3.2 Local neighborhood retention: NRS

Global subspace motion does not determine local neighbor retention. The NRS measurement family measures local retention directly by comparing nearest-neighbor sets for the same anchors in adjacent layers. We adapt the neighborhood-overlap idea of Valeriani et al. [68], instantiated here as cosine k_{NRS}-nearest-neighbor sets over the rows of each X_l.

Let k_{NRS} be the NRS neighborhood size, and let $\mathcal{A} \subseteq \{1, \ldots, N\}$ be a fixed random anchor set used across all layers, with $|\mathcal{A}| = \min(500, N)$ in our experiments. For anchor $i \in \mathcal{A}$, let $\mathcal{N}_l(i) = \mathrm{kNN}_{k_{\mathrm{NRS}}}(i; X_l)$ be the indices of the k_{NRS} nearest rows to row i of X_l under cosine distance, excluding i itself. The Jaccard retention between adjacent layers is

$$J_l(i) = \frac{|\mathcal{N}_l(i) \cap \mathcal{N}_{l+1}(i)|}{|\mathcal{N}_l(i) \cup \mathcal{N}_{l+1}(i)|}, \qquad J_l = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} J_l(i). \tag{3}$$

High J_l means that layer $l+1$ preserves the local neighborhoods induced by layer l. Low J_l means that many anchors change nearest-neighbor sets. The next measurement family shifts the reference from adjacent layers to the final layer.

3.3 Final-layer alignment: GFMI

The GFMI measurement family compares each layer with the final-layer representation and adapts the filtration idea from topological data analysis to avoid choosing a single graph threshold [12]. Let $G_l(\tau)$ denote the cosine k_{GFMI}-nearest-neighbor graph at layer l in which the edge between examples i and j is kept whenever its cosine distance is at most the τ-th percentile of layer l's edge-distance distribution. Because edge-distance scales differ across layers, the threshold is parameterized by per-layer percentiles rather than by shared absolute distances, and we use $\tau \in [0, 100]$ throughout. Small τ keeps only very close neighbors and produces fine-grained groups. Larger τ merges these groups into coarser components. We refer to this percentile sweep as a graph filtration. If an intermediate layer is already organized like the final layer, the two filtrations induce similar component partitions across many percentiles.

Let $C_l(\tau) = \mathrm{CC}(G_l(\tau))$ be the connected-component partition at layer l, and let $C_L(\tau)$ be the corresponding partition at the final layer. We compare the two partitions with mutual information [52], written $\mathrm{MI}_l(\tau) = I(C_l(\tau); C_L(\tau))$. This gives one value per percentile. A layer receives high values when its connected-component partition resembles the final-layer partition across many percentiles. We summarize the sweep by integrating $\mathrm{MI}_l(\tau)$ over the percentile range $[5, 95]$,

$$\mathrm{GFMI}(l) = \int_5^{95} \mathrm{MI}_l(\tau)\, d\tau, \tag{4}$$

implemented numerically by trapezoidal integration over a discrete percentile grid. We refer to this value as the GFMI area under the curve (GFMI AUC) in the result tables. Here $I(\cdot;\cdot)$ is raw discrete mutual information, so GFMI values are not bounded in $[0, 1]$, and the self-GFMI of the final layer reduces to the integrated entropy of $C_L(\tau)$ over the same percentile range.

The three measurement families above produce layer-wise sequences, and these sequences are kept separate throughout. Sections 5 and 6 apply two different reductions: model selection turns each sequence into one model-level scalar, and layer pruning uses per-layer scores. The exact reductions are defined in those sections.

4 Setup and Layer-wise Observations

We apply the measurements from Section 3 to 31 models and 30 MTEB tasks. For each model-task pair, we pool each layer's token-level hidden states into one vector per example using the official or model-recommended pooling rule when available, and compute the measurements over the per-layer outputs.

4.1 Models and tasks

The model pool contains 25 embedding models with public MTEB references and six base LLMs with MMLU references. The embedding models span two architectural sub-classes: *encoder-based embedders* built on bidirectional encoder backbones (e.g., E5 [72], BGE [8]) and *decoder-based embedders* built on causal LLM backbones (e.g., SFR-Embedding [45], Qwen3-Embedding [77], NV-Embed-v2 [35]). The base LLMs are generation-oriented models that have not been contrastively tuned for retrieval, including Mistral 7B [30] and Gemma 2 [63]. The complete model and task panels are provided in Appendix B.1. The 30 tasks cover classification, retrieval, Semantic Textual Similarity (STS), reranking, and duplicate detection. Examples include Banking77 [6], Natural Questions [33], SciFact [71], STS Benchmark [7], and MIND [74]. For base LLMs, we compute LRD on the same 30 MTEB input sets, but do not use MTEB as a downstream quality reference. MTEB therefore provides both inputs and task-level quality references for embedding models, while MMLU provides only a model-level reference for base LLMs.

4.2 Layer-wise observations by architecture family and task type

We summarize the per-layer sequences of Section 3 with derived summaries, introduced alongside the patterns they describe. These summaries are descriptive, intended to expose family- and task-level structure. The application-specific scores used in Sections 5 and 6 are defined separately in those sections. Table 1 reports family-level shape and trend summaries averaged over all tasks, while Table 2 reports magnitude summaries split by task type.

Architecture-level patterns. For curvature (Eq. 2), we use the normalized peak depth $d_{\mathrm{peak}}(\kappa) = (\arg\max_{l \in \mathcal{L}_\kappa} \kappa_l)/L_\kappa$, where $\mathcal{L}_\kappa$ is the set of valid curvature indices and $L_\kappa = \max \mathcal{L}_\kappa$. For NRS (Eq. 3), we use the late-vs-early trend

$$\Delta J = \frac{1}{|\mathcal{L}_{\mathrm{late}}|} \sum_{l \in \mathcal{L}_{\mathrm{late}}} J_l - \frac{1}{|\mathcal{L}_{\mathrm{early}}|} \sum_{l \in \mathcal{L}_{\mathrm{early}}} J_l, \tag{5}$$

where $\mathcal{L}_{\mathrm{early}}$ and $\mathcal{L}_{\mathrm{late}}$ denote the first and last thirds of the layer indices.

Curvature peak depth $d_{\mathrm{peak}}(\kappa)$ separates the three families: encoder embedders peak at mid-depth, whereas decoder embedders and base LLMs peak later (Table 1). NRS retention shows the same split. Encoder embedders are nearly flat ($\Delta J \approx 0.01$), while in decoder embedders and base LLMs the local neighborhoods stabilize toward later layers. Both signals are consistent with prior layer-wise analyses showing that bidirectional encoders concentrate processing in the middle [65, 69], whereas causal LMs accumulate information bottom-up.

Table 1: Layer-wise patterns by architecture family. Symbols are defined in the main text. Per-model averages are in Appendix C.

Family	Models	$d_{\mathrm{peak}}(\kappa)$	ΔJ
Encoder embedder	13	0.56	+0.015
Decoder embedder	12	0.89	+0.144
Base LLM	6	0.78	+0.121

Task-type patterns. For task-type analysis we use three layer-aggregate magnitude summaries: the mean curvature $\bar{\kappa} = |\mathcal{L}_\kappa|^{-1} \sum_{l \in \mathcal{L}_\kappa} \kappa_l$, the mean speed $\bar{s} = |\mathcal{L}_s|^{-1} \sum_{l \in \mathcal{L}_s} s_l$ (with $\mathcal{L}_s$ the set of valid speed indices), and the GFMI scale $\bar{G} = (L+1)^{-1} \sum_{l=0}^{L} \mathrm{GFMI}(l)$ (Eq. 4). Because raw GFMI is bounded by the partition entropy of the final-layer kNN graph, which itself differs across

families, we treat $\bar{G}$ as a within-family magnitude signal. The first two quantities capture trajectory curvature and per-layer displacement, respectively.

Table 2 shows a consistent split by task type: classification has higher mean curvature in the encoder and decoder families, with base LLMs showing little task-type difference, whereas retrieval has higher mean speed and GFMI scale. This pattern is consistent with task geometry. Classification organizes examples around discrete labels, which matches sharper trajectory curvature. Retrieval compares sample pairs by relative distance rather than discrete labels, which is consistent with transformations spreading across more layers and with the per-layer kNN graph aligning with the final-layer graph earlier. The latter is consistent with evidence that retrieval ability is encoded primarily in shallow layers [61].

Table 2: Task-type stratified layer-aggregate summaries by model family. Means are over model-task pairs. CLS = classification, RET = retrieval, Other = STS / reranking / pair detection.

	Mean curv. $\bar{\kappa}$			Mean speed $\bar{s}$			GFMI scale $\bar{G}$		
Family	CLS	RET	Other	CLS	RET	Other	CLS	RET	Other
Encoder embedder	0.137	0.132	0.142	12.299	12.749	11.931	93.4	98.1	109.8
Decoder embedder	0.142	0.132	0.133	11.873	12.632	12.263	247.9	258.8	247.5
Base LLM	0.346	0.345	0.255	7.695	8.024	8.873	325.9	351.7	314.6

5 Model Selection with LRD

Consider choosing among candidate embedding models for a deployment task whose downstream score is unavailable, e.g. a proprietary corpus without labeled queries. We test whether the layer-wise measurements of Section 3, computed only from task inputs, can rank models in agreement with downstream references on a 25-model MTEB panel and a 6-LLM MMLU panel. From each family we derive a single model-level selection score ($d_{0,L}$ for Frenet, $\bar{J}_{\text{late}}$ for NRS, $\overline{G}^{\text{peak}}$ for GFMI), each compressing the layer-wise sequence in a different way. Agreement across all three would indicate that the layer-wise view is not tied to a single aggregation choice.

5.1 Model-level selection scores

Frenet end-to-end distance: $d_{0,L}$. The Frenet selection score $d_{0,L} = d_{\text{Gr}}(Q_0, Q_L)$ (Eq. 1) is the Grassmann distance between the input and output subspaces, in contrast to the per-step distances $s_l = d_{l,l+1}$ that summarize adjacent-layer motion (Appendix A). It captures the net displacement of the representation subspace from input to output, and unlike path-integrated Frenet derivatives does not scale with the layer count L, so it remains comparable across architectures of different depths. Embedding models whose hidden subspace moves further between the input and the final layer are expected to produce stronger downstream representations.

NRS late-layer retention: $\bar{J}_{\text{late}}$. The NRS selection score $\bar{J}_{\text{late}}$ is the mean Jaccard retention J_l (Eq. 3) between adjacent layers over the late third of adjacent-layer transitions (Appendix A). It measures how stable the local neighborhood structure becomes in the layers closest to the final layer. We restrict to the late third because these layers carry the representations exposed to downstream tasks, and because Section 4 shows that late-layer NRS retention differs systematically across model families. Embedding models whose late layers stop substantially rewiring local sample neighborhoods are expected to yield stronger downstream representations.

GFMI peak alignment: $\overline{G}^{\text{peak}}$. The GFMI selection score $\overline{G}^{\text{peak}}$ averages, over all $L+1$ layers including the final layer, $\sup_\tau \text{MI}_l(\tau)$, where $\text{MI}_l(\tau)$ is the per-percentile mutual information defined in Section 3 (Appendix A). The per-layer maximization over τ lets each layer use the percentile at which $\text{MI}_l(\tau)$ is largest, rather than forcing all layers to share one fixed percentile. Embedding models whose intermediate layers are already well aligned with the final graph organization are expected to produce stronger downstream representations.

Table 3: Label-free model-selection correlations. MTEB entries are cross-task mean±sd Spearman correlations on the 25-model embedding panel. Parentheses give the number of significant tasks. MMLU is a 6-model directional check.

		MTEB-25				MMLU-6
Metric	Family	All (sig./30)	CLS (13)	RET (11)	Other (6)	ρ
$d_{0,L}$	Frenet	$+0.64 \pm 0.17$ (28)	$+0.67 \pm 0.13$	$+0.68 \pm 0.16$	$+0.50 \pm 0.22$	$+0.64$
$\bar{J}_{\text{late}}$	NRS	$+0.58 \pm 0.17$ (26)	$+0.64 \pm 0.11$	$+0.57 \pm 0.17$	$+0.49 \pm 0.26$	$+0.46$
$\overline{G}^{\text{peak}}$	GFMI	$+0.52 \pm 0.19$ (23)	$+0.60 \pm 0.14$	$+0.51 \pm 0.09$	$+0.36 \pm 0.32$	$+0.64$

5.2 Setup

Both panels use the same 30 MTEB task inputs as the source of layer-wise representations. What differs is the downstream quality reference and the model class being tested. The MTEB panel is our primary evidence: 25 embedding models with 30 task-specific MTEB scores [48]. For each task we compute the Spearman correlation between the selection score and the MTEB score across the 25 models, and report the cross-task mean and standard deviation together with the number of tasks reaching $p < 0.05$. The MMLU panel tests whether the same scores generalize to a different model class and benchmark. It comprises 6 base LLMs, each with a single aggregate MMLU score [24]. We average each LLM's selection score over the 30 task inputs and compute one Spearman correlation across the 6 LLMs. With $n = 6$ this panel resolves only large effects, so we report it as a directional generalization check.

5.3 Results

Table 3 reports the MTEB-25 results: $d_{0,L}$ is the strongest selection score with the highest mean Spearman correlation ($+0.64$) and significance on 28 of 30 tasks. $\bar{J}_{\text{late}}$ ($+0.58$) and $\overline{G}^{\text{peak}}$ ($+0.52$) are positive but weaker. Stratifying by task type (Table 3, right side) reveals a task-type split. On classification, all three scores remain close ($+0.67$ / $+0.64$ / $+0.60$), suggesting that each measurement family carries comparable selection information. On retrieval, $d_{0,L}$ remains high at $+0.68$, while $\bar{J}_{\text{late}}$ and $\overline{G}^{\text{peak}}$ fall to $+0.57$ and $+0.51$. The weaker GFMI signal on retrieval is consistent with its graph-partition target: classification can produce more discrete final-layer clusters, whereas retrieval and similarity-style tasks depend more on continuous neighborhood geometry. Mean correlations are lower on the smaller *Other* group (STS, reranking, pair / duplicate detection: 6 tasks, 2 per subtype), where the cross-task mean is dominated by within-subtype variation. Spearman captures global rank agreement, and top-k selection quality is left to future work. Appendix D.1 reports the corresponding task-level correlations.

Figure 1 shows the model-level view: for each score, the x-axis is its mean over the 30 task inputs and the y-axis is the model's mean official MTEB score over those same 30 tasks. A monotone increasing trend indicates that the label-free score and the labeled benchmark agree at the model level. The trend is clearest for $d_{0,L}$ and consistently positive for the other two scores, so a label-free pre-selection using any of them prefers higher-MTEB models on average.

Averaging each LLM's selection score over the 30 task inputs and correlating against MMLU gives Spearman $\rho = +0.64$ for both $d_{0,L}$ and $\overline{G}^{\text{peak}}$, and $+0.46$ for $\bar{J}_{\text{late}}$ (Table 3). All three correlations remain positive on MMLU, indicating that the layer-wise score does not flip sign across the embedder/base-LLM boundary. With $n = 6$, this is a directional generalization check, and we do not interpret the relative ordering of the three scores on this panel.

6 Layer Pruning with LRD

We next test whether the same layer-wise measurements carry layer-level redundancy signal. For a fixed model and task, we rank layers by a measurement-derived per-layer score, skip the corresponding transformer blocks at inference time without retraining or recovery, and report the score change relative to the unpruned model. Layers flagged as redundant by an informative measurement should cause smaller score drops than layers selected by random or last-k baselines.

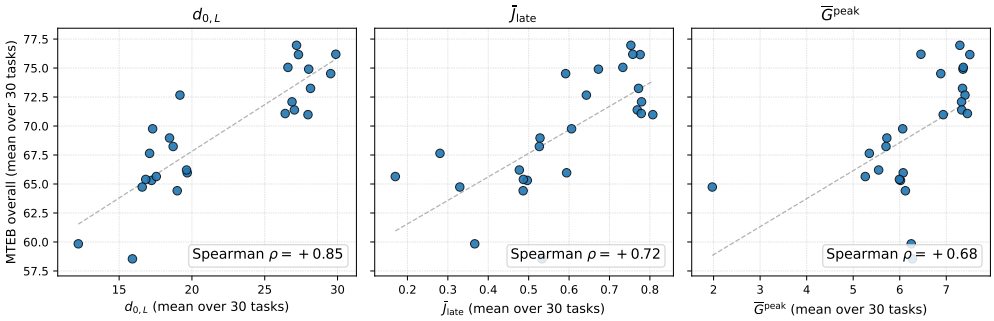


Figure 1: Per-model selection-score scatter on the MTEB-25 panel. Scores and official MTEB results are averaged over the 30 tasks, with Spearman ρ annotated in each panel.

6.1 Pruning rules

For each model and task we convert each family's layer-wise sequence into a per-layer pruning score by independently z-scoring within that sequence (Appendix A). The Frenet rule combines low Grassmann speed (Eq. 1) with low Menger curvature (Eq. 2). The NRS rule prefers layers with high Jaccard overlap to the previous layer. The GFMI rule prefers layers with high filtration AUC against the final graph. A higher score means the layer contributes less under its family's measurement and is removed first at each pruning budget: in Frenet terms the subspace barely moves or bends, in NRS terms the local neighborhoods are already stable, and in GFMI terms the layer is already close to the final graph. The GFMI criterion is indirect: it does not measure the block's local transformation size but whether the layer's graph organization already resembles the final-layer pattern. Compared to the model-level scores of Section 5, here we use each layer's measurement directly rather than aggregating to a single scalar. For GFMI this means using $\mathrm{GFMI}(l)$ from Section 3 (Eq. 4) at each layer, rather than the $\sup_\tau \mathrm{MI}_l(\tau)$ summary used in $\overline{G}^{\mathrm{peak}}$ (Section 5).

6.2 Setup

We evaluate on three MTEB tasks (AmazonPolarityClassification, FiQA2018, STSBenchmark) and five models from the three architectural families: two encoder embedders, two decoder embedders, and one base LLM.[1]

Each (model, task) cell runs the three rules of Appendix A and two baselines: random removal averaged over 3 seeds and last-k removal (the k deepest unprotected layers). The first three and last three layers are protected. Pruning budgets of $\{5, 10, 15, 20\}\%$ are taken over the remaining layers. We report $\mathrm{rel}\Delta = 100\,(s_{\mathrm{pruned}} - s_{\mathrm{unpruned}})/s_{\mathrm{unpruned}}$, where s is the official MTEB metric for that task. Values closer to zero indicate less degradation.

6.3 Results

Table 4: Layer-pruning relative score change across budgets, aggregated over 15 model-task cells. Values closer to 0 are better.

	Mean rel. Δ (%)				Median rel. Δ (%)			
Rule	5%	10%	15%	20%	5%	10%	15%	20%
Random	−4.79	**−4.14**	−5.90	−12.08	−1.96	−2.88	−5.41	−5.41
Last-k	−6.73	−11.76	−14.76	−13.13	−1.94	−5.47	−8.66	−8.66
Frenet	**−2.10**	−8.83	−19.62	−21.33	−0.98	−5.32	−15.12	−14.19
NRS	−9.54	−14.90	−15.05	−15.50	−1.94	−5.51	−7.27	−9.89
GFMI	−5.09	−6.95	**−2.37**	**−6.14**	**−0.20**	**−0.20**	**−1.31**	**−3.27**

[1]The five models are BGE-base-en-v1.5, E5-large-v2, Nemotron-Embed-1B, LLM2Vec-Mistral-7B, and Mistral-7B-v0.1.

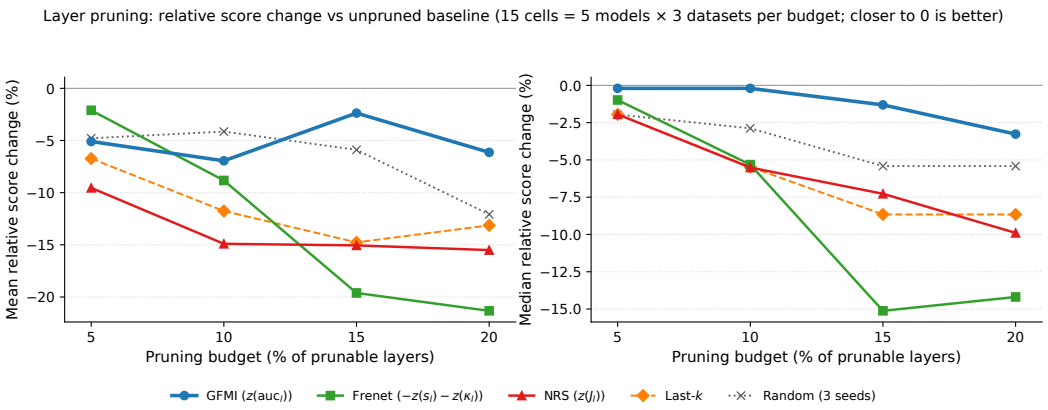


Figure 2: Layer-pruning score change over 15 model-task cells. GFMI improves over Random by mean at 15% and 20%, and is closest to zero by median at every budget.

Table 4 and Figure 2 support three observations. First, GFMI gives the clearest advantage at the larger pruning budgets: by mean, it improves over Random at 15% (-2.37% vs -5.90%) and 20% (-6.14% vs -12.08%). At lighter budgets, rule ordering is less consistent: Frenet has the best mean at 5%, Random has the best mean at 10%, and GFMI separates most clearly in the median panel. The median panel is less sensitive to catastrophic cells and shows GFMI closest to zero at every budget. Second, the Frenet rule is competitive at the lightest budget (-2.10% mean at 5%, the lowest of any rule) but degrades rapidly as the budget grows (-21.33% at 20%). This makes the Frenet rule a light-budget signal rather than a stable rule for larger pruning budgets. Third, the NRS rule is the weakest of the three: although $\bar{J}_{\text{late}}$ is a positive model-level selection signal in Section 5, the per-layer rule $z(J_l)$ does not carry over to pruning, since a layer that does not change local neighborhoods is not necessarily dispensable.

The Random baseline is itself competitive at light budgets, in line with prior layer-removal work [14, 44, 21]. The GFMI advantage is concentrated at higher budgets and is largest on encoder embedders. On LLMs the three rules are harder to separate from Random. Appendix D.2 reports the aggregate mean-curve view and full per-cell pruning results.

7 Limitations and Conclusion

LRD comprises three layer-wise measurement families (Frenet, NRS, GFMI) for global subspace motion, local neighborhood stability, and final-layer alignment. On 31 models and 30 MTEB tasks, it exposes architecture- and task-level patterns (Section 4). Its three model-level selection scores correlate positively with downstream MTEB performance (Section 5), and GFMI beats Random at the 15% and 20% pruning budgets (Section 6). The NRS results show that model-level and layer-level signals need not transfer: $\bar{J}_{\text{late}}$ is positive for selection, while $z(J_l)$ is the weakest pruning rule.

Limitations remain. Model selection relies mainly on the 25-model MTEB embedding panel, with the 6-LLM MMLU result only a directional generalization check ($n = 6$). The embedding panel uses public MTEB models and should be tested on less curated pools. We omit head-to-head label-free or transferability baselines, while Section 2 gives the conceptual contrast with single-layer summaries. The selection study also evaluates global rank agreement, leaving task-specific top-k selection behavior for a larger benchmark. The pruning panel has five models, three tasks, and one base LLM, and uses inference-time block skipping without retraining, so it is not a compression benchmark. It therefore isolates whether LRD scores identify less damaging layers, not whether a compressed model can be recovered after pruning. Random is competitive at light budgets, and the measurement-guided prior is supported only for GFMI past the 10% budget. The measurements also depend on input sampling, pooling, k-NN construction, and graph filtration percentile. Broader task coverage and stability analyses remain open.

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A Formula Details and Conventions

A.1 Primitive conventions

This subsection records the finite-sample conventions needed to instantiate the primitives defined in Section 3. It supplements the main text by fixing indexing, boundary cases, and discretization choices.

Frenet. For a fixed model and task, we use the final-layer rank convention from Section 3 for all layers. Speed is forward indexed: $s_l = d_{\mathrm{Gr}}(Q_l, Q_{l+1})$ for $l = 0, \ldots, L-1$. Curvature is assigned to the middle layer of each consecutive triple, so κ_l is computed from (Q_{l-1}, Q_l, Q_{l+1}) for $l = 1, \ldots, L-1$. Degenerate or zero-area triples receive curvature zero.

NRS. The anchor set is sampled once for a fixed model and task and reused across all layers. It contains up to 500 examples, sampled without replacement from the task inputs. We use cosine k_{NRS}-nearest neighbors with $k_{\mathrm{NRS}} = 20$. Thus J_l compares layers l and $l+1$ and is defined for $l = 0, \ldots, L-1$.

GFMI. GFMI uses cosine k_{GFMI}-nearest-neighbor candidate graphs with $k_{\mathrm{GFMI}} = 30$. The filtration parameter τ is a per-layer percentile rather than a shared raw cosine-distance threshold, so different layers can sweep different absolute distance ranges. The thresholded graph is symmetrized before connected components are computed. Both the integrated GFMI value and the peak summary are evaluated on the same discrete percentile grid over $[5, 95]$. The integrated value is the trapezoidal area under the per-percentile MI curve, with no additional normalization.

A.2 Model-level selection scores

The selection scores in Section 5 reduce each measurement family to a single scalar for each fixed model and task. They are deliberately simple reductions of the primitives in Section 3, so that the model-selection experiment tests whether the layer-wise signals survive a direct aggregation rather than a task-specific fitting procedure.

Frenet end-to-end distance. The Frenet score uses the end-to-end subspace displacement rather than a path-integrated quantity. This makes the score comparable across models of different depths while still measuring how far the representation subspace moves from the input side to the final layer. With Q_0, Q_L the orthonormal PCA bases at layers 0 and L,

$$d_{0,L} = d_{\mathrm{Gr}}(Q_0, Q_L). \tag{6}$$

NRS late-layer retention. The NRS score focuses on the late third of adjacent-layer transitions, because these transitions are closest to the representations exposed to downstream tasks and Section 4 shows that late-layer retention differs systematically across model families. Let $\mathcal{L}_{\mathrm{late}} = \{\lceil 2L/3 \rceil, \ldots, L-1\}$ index the late third of adjacent-layer transitions. The NRS selection score is

$$\bar{J}_{\mathrm{late}} \;=\; \frac{1}{|\mathcal{L}_{\mathrm{late}}|} \sum_{l \in \mathcal{L}_{\mathrm{late}}} J_l. \tag{7}$$

GFMI peak alignment. The GFMI score uses a per-layer peak before averaging over layers. This lets each layer contribute at the percentile where its component partition best matches the final-layer partition, rather than forcing all layers to share one fixed percentile. The GFMI selection score is

$$\overline{G}^{\mathrm{peak}} \;=\; \frac{1}{L+1} \sum_{l=0}^{L} \sup_{\tau \in \mathcal{T}} \mathrm{MI}_l(\tau). \tag{8}$$

A.3 Per-layer pruning rules

The pruning rules in Section 6 turn each primitive sequence into a layer-level redundancy score. The scores are standardized within each fixed model and task, using

$$z(x_l) = \frac{x_l - \mu_x}{\sigma_x}, \tag{9}$$

where μ_x and σ_x are computed over the valid indices of the corresponding primitive sequence. The three rules are

$$\begin{aligned} \text{frenet}_l &= -z(s_l) - z(\kappa_l), \\ \text{nrs}_l &= z(J_l), \\ \text{gfmi}_l &= z(\text{GFMI}(l)). \end{aligned} \tag{10}$$

The signs make larger scores mean more removable: Frenet favors layers with low speed and low curvature, NRS favors layers that preserve adjacent-layer neighborhoods, and GFMI favors layers already aligned with the final-layer graph. Layers are sorted in decreasing order of the rule score and removed up to the budget. The first three and last three layers of every model are excluded from removal, which also covers boundary indices where a primitive is undefined, such as s_l at $l = L$, κ_l at $l = 0$ or $l = L$, and J_l at $l = L$.

B Experimental Details

B.1 Models and tasks

Tables B.1 and B.2 list the 31-model and 30-task panel used for the main layer-wise analysis and model-selection study.

Table B.1: Model panel used for LRD measurements. "#L" denotes transformer layers.

Model	Family / Backbone	#L
Encoder embedders (13)		
all-MiniLM-L6-v2 [56]	MiniLM (BERT-style)	6
all-mpnet-base-v2 [56]	MPNet (BERT-style)	12
BGE-base-en-v1.5 [8]	BERT-base	12
BGE-large-en-v1.5 [8]	BERT-large	24
E5-base-v2 [72]	BERT-base	12
E5-large-v2 [72]	BERT-large	24
GTE-base [38]	BERT-base	12
GTE-large [38]	BERT-large	24
Jina-Embeddings-v3 [62]	XLM-RoBERTa	24
Mxbai-Embed-Large-v1	BERT-large	24
Nomic-Embed-Text-v1.5 [50]	Nomic BERT	12
Snowflake-Arctic-Embed-L [46]	BERT-large	24
Snowflake-Arctic-Embed-M-v2 [76]	GTE-multilingual	12
Decoder embedders (12)		
BGE-en-icl [8]	Mistral-7B	32
BGE-multilingual-gemma2 [8]	Gemma-2-9B	42
E5-Mistral-7B-Instruct [72]	Mistral-7B	32
GritLM-7B [49]	Mistral-7B	32
GTE-Qwen2-7B-Instruct [38]	Qwen2-7B	28
Linq-Embed-Mistral	Mistral-7B	32
NV-Embed-v2 [35]	Mistral-7B	32
Qwen3-Embedding-0.6B [77]	Qwen3-0.6B	28
Qwen3-Embedding-8B [77]	Qwen3-8B	36
SFR-Embedding-2-R [45]	Mistral-7B	32
SFR-Embedding-Mistral [45]	Mistral-7B	32
Zeta-Alpha-E5-Mistral	Mistral-7B	32
Base LLMs (6)		
Falcon-7B [2]	Falcon	32
Gemma-2-2B [63]	Gemma-2	26
GPT-2 [53]	GPT-2	12
GPT-2 XL [53]	GPT-2	48
Llama-3.2-1B [20]	Llama-3	16
Mistral-7B-v0.1 [30]	Mistral-7B	32

Table B.2: MTEB input sets used for LRD measurements. Metric is the official MTEB main metric [48].

Task	Type	Metric
Classification (13)		
AmazonCounterfactualClassification [51]	Classification	Accuracy
AmazonPolarityClassification [43]	Classification	Accuracy
Banking77Classification [6]	Classification	Accuracy
DBpediaClassification [23]	Classification	Accuracy
EmotionClassification [58]	Classification	Accuracy
ImdbClassification [41]	Classification	Accuracy
MassiveIntentClassification [16]	Classification	Accuracy
MassiveScenarioClassification [16]	Classification	Accuracy
MTOPDomainClassification [36]	Classification	Accuracy
MTOPIntentClassification [36]	Classification	Accuracy
ToxicConversationsClassification	Classification	Accuracy
TweetSentimentExtractionClassification	Classification	Accuracy
TweetTopicSingleClassification	Classification	Accuracy
Retrieval (11)		
ArguAna [70]	Retrieval	nDCG@10
CQADupstackRetrieval [26]	Retrieval	nDCG@10
FiQA2018 [66]	Retrieval	nDCG@10
HotpotQA [75]	Retrieval	nDCG@10
NFCorpus [5]	Retrieval	nDCG@10
NQ [33]	Retrieval	nDCG@10
QuoraRetrieval	Retrieval	nDCG@10
SCIDOCS [9]	Retrieval	nDCG@10
SciFact [71]	Retrieval	nDCG@10
Touche2020	Retrieval	nDCG@10
TRECCOVID [57]	Retrieval	nDCG@10
Other (6)		
AskUbuntuDupQuestions	Reranking	MAP
MindSmallReranking [74]	Reranking	MAP
SICK-R [42]	STS	Spearman
STSBenchmark [7]	STS	Spearman
SprintDuplicateQuestions [59]	Duplicate detection	AP
TwitterURLCorpus [34]	Duplicate detection	AP

B.2 Protocols and compute

LRD measurements are computed from forward-pass hidden states without training or fine-tuning. For each model-task pair, the task inputs define a layer-wise representation sequence $X_0, \ldots, X_L$, and the same task inputs are reused across models within a measurement family. All measurement hyperparameters are fixed across experiments: Frenet uses the 95% final-layer PCA variance rule, NRS uses $k_{\mathrm{NRS}} = 20$, and GFMI uses $k_{\mathrm{GFMI}} = 30$ with 20 percentile points over $[5, 95]$. For model selection, we compute the Spearman correlation between each LRD selection score and the official MTEB score across the 25 embedding models for each task, then report the mean and standard deviation across the 30 tasks. For the MMLU directional check, we average each base LLM's LRD selection score over the 30 MTEB input sets and compute one Spearman correlation across the six base LLMs against their aggregate MMLU scores. For layer pruning, we evaluate five models on three tasks under pruning budgets $\{5, 10, 15, 20\}\%$ using the model and task panel listed in Section 6.2. We compare Random, last-k, Frenet, NRS, and GFMI rules. Random is averaged over three draws, and the first three and last three layers are protected for every rule.

Compute. Experiments were run on a server with four NVIDIA Tesla V100 GPUs (32GB each). The reported measurement and pruning experiments required approximately 300 GPU-hours in total.

C Additional Layer-wise Analyses

Section 4 reports family-level and task-type summaries computed from the raw layer-wise measurement sequences. The table below gives the per-model averages behind those summaries, with each row averaged over all 30 tasks. Layer-aggregate magnitudes match Section 4: $\bar{\kappa}$ is the average per-layer Menger curvature, $\bar{s}$ is the average Grassmann speed, and $\bar{G}$ is the average layer-wise integrated GFMI.

Figure C.1 separates broad family effects from high-leverage model effects. The position and trend summaries support the architecture-level pattern from Section 4: decoder embedders concentrate at late curvature peaks and positive NRS ΔJ, whereas encoder embedders show more dispersed peak locations and nearly flat NRS change. This agreement indicates that the same late-layer organization appears in both a global subspace statistic and a local neighborhood statistic. The magnitude summaries distinguish base LLMs from decoder embedders rather than placing them on the same continuum. Base LLMs have higher mean curvature and GFMI scale, lower mean speed, and substantial within-family spread, so their family averages should be read together with the per-model dispersion.

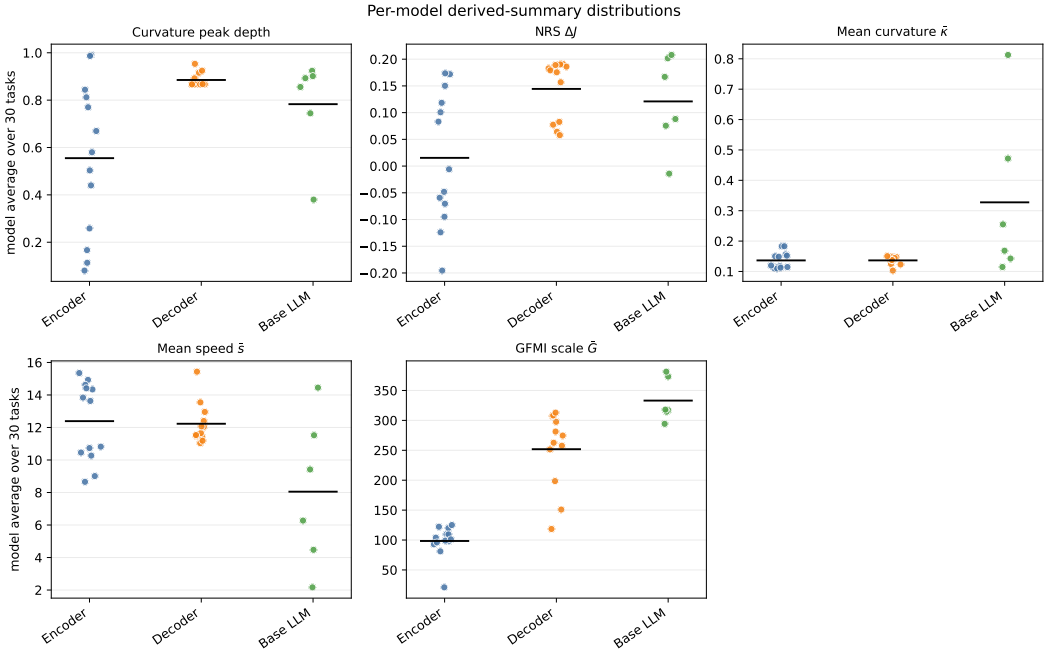


Figure C.1: Per-model distributions of the five layer-wise summaries used in Section 4. Dots are task-averaged model values and black bars mark family means.

Figure C.2 shows that the summaries are structured but not interchangeable. In the left panel, many decoder embedders fall in the late-peak and positive-ΔJ region, linking late global subspace bending with late local-neighborhood stabilization. Encoders occupy a broader range of peak depths but stay closer to flat or negative ΔJ, consistent with weaker late-layer neighborhood stabilization. The middle panel separates GFMI from speed: encoders and decoders can have overlapping mean speed, yet encoders occupy a lower-GFMI band. Final-layer graph alignment therefore captures structure beyond subspace displacement. The right panel gives the clearest base-LLM separation. Encoders and decoders mostly remain in a low-curvature, higher-speed region, whereas several base models move toward higher curvature and lower speed. Together these panels support the use of three complementary summaries rather than a single aggregate score.

Table C.3 reports the exact per-model values used by Figure C.1. These values indicate whether a family-level effect is distributed across many models or concentrated in a small number of high-leverage cases. They clarify three patterns that are only visible qualitatively in the figures. First, encoder embedders have the widest spread in curvature peak depth, ranging from early E5 peaks to

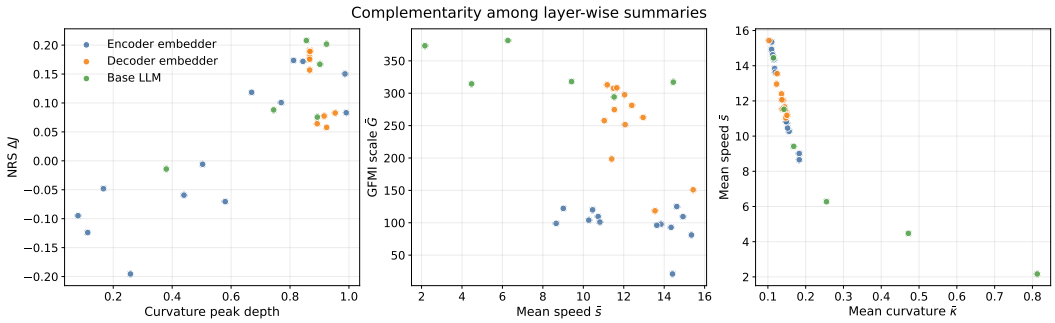


Figure C.2: Pairwise views of complementary layer-wise summaries. Each point is one model averaged over the 30-task panel.

late Snowflake peaks. Second, decoder embedders are more concentrated in curvature peak depth and NRS ΔJ, but their GFMI values still vary substantially, with NV-Embed-v2 and GTE-Qwen2-7B-Instruct below the main decoder cluster. Third, base LLMs show the largest magnitude heterogeneity: GPT-2 has high mean curvature and low mean speed, while Llama-3.2-1B and Mistral-7B-v0.1 are closer to the decoder speed range.

Table C.3: Per-model task-averaged summaries over the 30-task panel.

Model	Curv. peak	ΔJ	$\bar{\kappa}$	$\bar{s}$	$\bar{G}$
Encoder embedders					
all-MiniLM-L6-v2	0.258	-0.196	0.183	9.010	122.2
all-mpnet-base-v2	0.440	-0.059	0.150	10.733	109.6
BGE-base-en-v1.5	0.770	0.101	0.116	14.342	92.8
BGE-large-en-v1.5	0.844	0.172	0.118	13.841	97.8
E5-base-v2	0.113	-0.124	0.156	10.270	104.0
E5-large-v2	0.080	-0.095	0.152	10.459	119.9
GTE-base	0.503	-0.006	0.110	15.349	81.1
GTE-large	0.670	0.118	0.110	14.929	109.5
Jina-Embeddings-v3	0.167	-0.048	0.183	8.657	99.0
Mxbai-Embed-Large-v1	0.812	0.174	0.120	13.634	96.1
Nomic-Embed-Text-v1.5	0.580	-0.070	0.149	10.820	101.1
Snowflake-Arctic-Embed-L	0.991	0.083	0.113	14.622	125.1
Snowflake-Arctic-Embed-M-v2	0.987	0.150	0.115	14.411	20.9
Decoder embedders					
BGE-en-icl	0.867	0.176	0.137	12.068	251.5
BGE-multilingual-gemma2	0.953	0.083	0.138	11.534	274.6
E5-Mistral-7B-Instruct	0.868	0.189	0.150	11.184	313.0
GritLM-7B	0.868	0.191	0.147	11.403	198.5
GTE-Qwen2-7B-Instruct	0.924	0.058	0.103	15.433	150.9
Linq-Embed-Mistral	0.867	0.183	0.136	12.399	281.3
NV-Embed-v2	0.867	0.157	0.125	13.546	118.4
Qwen3-Embedding-0.6B	0.892	0.064	0.147	11.036	257.6
Qwen3-Embedding-8B	0.916	0.077	0.123	12.957	262.6
SFR-Embedding-2-R	0.867	0.186	0.140	12.042	297.6
SFR-Embedding-Mistral	0.867	0.190	0.146	11.492	307.6
Zeta-Alpha-E5-Mistral	0.867	0.179	0.144	11.658	308.4
Base LLMs					
Falcon-7B	0.744	0.088	0.472	4.474	314.4
Gemma-2-2B	0.901	0.167	0.168	9.418	318.0
GPT-2	0.380	-0.014	0.813	2.166	373.3
GPT-2 XL	0.893	0.076	0.255	6.273	381.4

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Model	Curv. peak	ΔJ	$\bar{\kappa}$	$\bar{s}$	$\bar{G}$
Llama-3.2-1B	0.924	0.202	0.115	14.451	317.1
Mistral-7B-v0.1	0.856	0.208	0.143	11.523	294.2

D Application-level supporting results

This section provides a disaggregated view of the two application studies. It does not introduce additional claims beyond Sections 5 and 6. Instead, it reports cell-level evidence underlying the aggregate results in Tables 3 and 4. The model-selection figures and tables examine whether the positive mean Spearman correlations in Table 3 are broadly distributed across tasks, while the pruning figure and table identify which cells account for the mean and median patterns in Table 4.

D.1 Model selection

Figure D.3 and Table D.4 report the dataset-conditioned model-selection correlations. Each row fixes one MTEB task, treats the 25 embedding models as candidates, and correlates an LRD score with the official MTEB score across those candidates, matching the evaluation axis used in Section 5. A high ρ means that the label-free score ranks stronger models higher for that task. Over the full panel, the Frenet end-to-end distance $d_{0,L}$ is the most stable signal. It is positive on every task and has the highest overall mean correlation ($+0.64$). The three families are closest on classification, where the task-type means are $+0.67$ for $d_{0,L}$, $+0.64$ for NRS, and $+0.60$ for GFMI. This supports the interpretation that local-neighborhood retention and graph-partition alignment are informative when the task is organized around discrete labels. The separation is larger on retrieval: $d_{0,L}$ remains high at $+0.68$, while NRS falls to $+0.57$ and GFMI to $+0.51$. These results position GFMI as a complementary model-selection signal rather than the primary criterion. Its graph-partition target has lower contrast when the final representation is not naturally clustered. The weakest cells are concentrated in the smaller Other group, where the means are $+0.50$, $+0.49$, and $+0.36$. GFMI is weak on MindSmallReranking ($+0.04$) and negative on SprintDuplicateQuestions (-0.07), so the low Other mean reflects identifiable task-level failures rather than only variance from averaging.

Table D.4: Task-level model-selection correlations. Entries report ρ with p-values in parentheses.

Task	$d_{0,L}$	$\bar{J}_{\text{late}}$	$\overline{G}^{\text{peak}}$
Classification			
AmazonCounterfactual	$+0.62$ ($< .001$)	$+0.61$ (0.001)	$+0.57$ (0.003)
AmazonPolarity	$+0.87$ ($< .001$)	$+0.78$ ($< .001$)	$+0.73$ ($< .001$)
Banking77	$+0.55$ (0.004)	$+0.43$ (0.032)	$+0.37$ (0.071)
DBpedia	$+0.53$ (0.011)	$+0.61$ (0.003)	$+0.57$ (0.006)
Emotion	$+0.67$ ($< .001$)	$+0.65$ ($< .001$)	$+0.56$ (0.004)
Imdb	$+0.84$ ($< .001$)	$+0.78$ ($< .001$)	$+0.75$ ($< .001$)
MTOPDomain	$+0.81$ ($< .001$)	$+0.65$ ($< .001$)	$+0.66$ ($< .001$)
MTOPIntent	$+0.59$ (0.002)	$+0.64$ ($< .001$)	$+0.60$ (0.002)
MassiveIntent	$+0.73$ ($< .001$)	$+0.73$ ($< .001$)	$+0.79$ ($< .001$)
MassiveScenario	$+0.63$ ($< .001$)	$+0.67$ ($< .001$)	$+0.68$ ($< .001$)
ToxicConversations	$+0.69$ ($< .001$)	$+0.60$ (0.001)	$+0.37$ (0.068)
TweetSentimentExtraction	$+0.76$ ($< .001$)	$+0.73$ ($< .001$)	$+0.69$ ($< .001$)
TweetTopicSingle	$+0.43$ (0.048)	$+0.41$ (0.059)	$+0.40$ (0.062)
Retrieval			
ArguAna	$+0.56$ (0.004)	$+0.46$ (0.021)	$+0.47$ (0.018)
CQADupstack	$+0.60$ (0.001)	$+0.62$ ($< .001$)	$+0.49$ (0.014)
FiQA2018	$+0.85$ ($< .001$)	$+0.80$ ($< .001$)	$+0.65$ ($< .001$)
HotpotQA	$+0.81$ ($< .001$)	$+0.66$ ($< .001$)	$+0.54$ (0.006)
NFCorpus	$+0.88$ ($< .001$)	$+0.73$ ($< .001$)	$+0.54$ (0.005)

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Task	$d_{0,L}$	$\bar{J}_{\text{late}}$	$\overline{G}^{\text{peak}}$
NQ	$+0.77$ ($< .001$)	$+0.72$ ($< .001$)	$+0.63$ ($< .001$)
Quora	$+0.66$ ($< .001$)	$+0.57$ (0.003)	$+0.54$ (0.006)
SCIDOCS	$+0.33$ (0.105)	$+0.21$ (0.310)	$+0.38$ (0.060)
SciFact	$+0.80$ ($< .001$)	$+0.59$ (0.002)	$+0.43$ (0.034)
TRECCOVID	$+0.70$ ($< .001$)	$+0.49$ (0.013)	$+0.43$ (0.031)
Touche2020	$+0.56$ (0.004)	$+0.40$ (0.048)	$+0.45$ (0.023)
Other			
AskUbuntuDupQuestions	$+0.66$ ($< .001$)	$+0.75$ ($< .001$)	$+0.70$ ($< .001$)
MindSmallReranking	$+0.47$ (0.018)	$+0.21$ (0.306)	$+0.04$ (0.852)
SICK-R	$+0.44$ (0.030)	$+0.59$ (0.002)	$+0.57$ (0.003)
STSBenchmark	$+0.56$ (0.004)	$+0.43$ (0.032)	$+0.32$ (0.113)
SprintDuplicateQuestions	$+0.12$ (0.558)	$+0.17$ (0.408)	-0.07 (0.748)
TwitterURLCorpus	$+0.77$ ($< .001$)	$+0.79$ ($< .001$)	$+0.62$ ($< .001$)

Figure D.4 is the task-type distribution view of the per-task correlations in Table D.4. Classification forms the most compact positive band, which is consistent with discrete-label objectives giving all three primitives a similar ranking signal. Better models tend to produce clearer class structure, so subspace displacement, local-neighborhood retention, and graph-partition alignment move together. Retrieval remains positive but spreads more widely, especially for NRS and GFMI. This broader spread is consistent with retrieval depending on continuous neighborhood geometry rather than a small number of discrete clusters, making local overlap and graph partitions more dataset-specific. The Other group is the least coherent aggregation because its six tasks mix reranking, STS, and duplicate detection. Its wider spread should therefore be read as task heterogeneity rather than as a stable estimate of one task family.

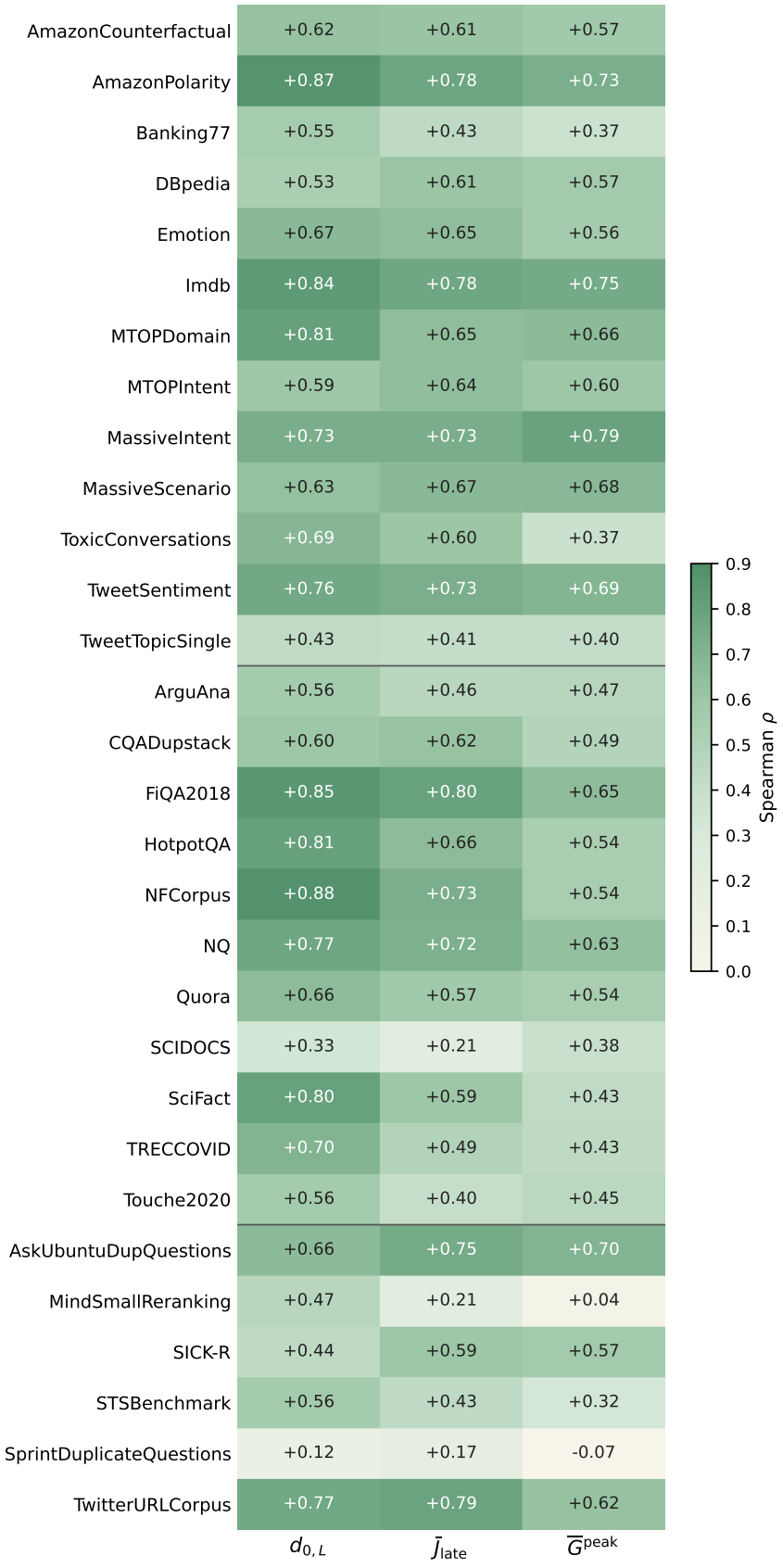


Figure D.3: Task-level Spearman correlations on the MTEB task panel.

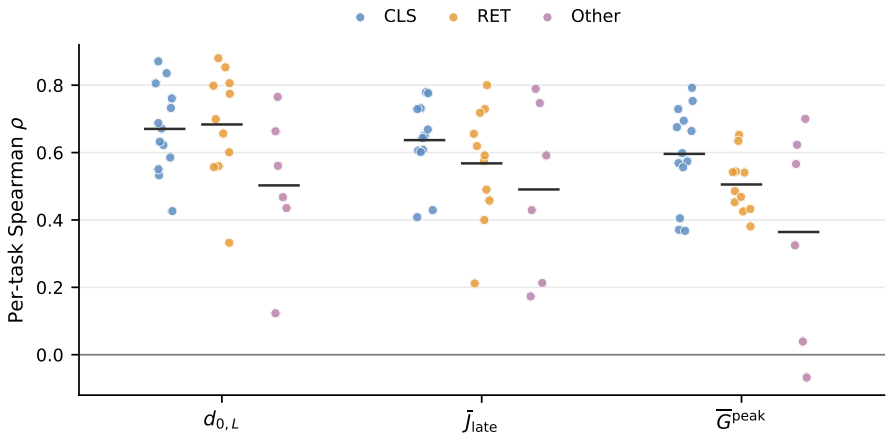


Figure D.4: Task-type distribution of per-task Spearman correlations.

Table D.5 reports the six-model MMLU panel used for the directional check in Section 5. Unlike the MTEB panel, this comparison has one downstream score per model rather than one score per task, so it tests whether task-averaged LRD scores preserve a coarse model-level ordering across base LLMs. The MMLU panel preserves the main ordering signal observed on MTEB: $d_{0,L}$ and $\overline{G}^{\text{peak}}$ both reach Spearman $\rho = +0.64$, whereas $\bar{J}_{\text{late}}$ is weaker at $\rho = +0.46$. Thus the two strongest MTEB-side scores also give the clearest model-level signal on the base-LLM panel.

Table D.5: Base-LLM model-selection panel. LRD scores are averaged over the 30 MTEB input sets before correlating with aggregate MMLU.

Model	MMLU	$d_{0,L}$	$\bar{J}_{\text{late}}$	$\overline{G}^{\text{peak}}$
Mistral-7B-v0.1	64.1	26.39	0.806	7.50
Gemma-2-2B	51.3	19.63	0.691	7.66
Llama-3.2-1B	32.2	21.44	0.684	7.61
Falcon-7B	27.8	11.67	0.557	6.21
GPT-2	26.0	4.71	0.577	7.19
GPT-2 XL	26.0	20.11	0.784	7.17

D.2 Layer pruning

Figure D.5 plots the mean relative score change from Table 4 as a function of pruning budget, averaging over the 15 model-task cells. The rule ordering changes with budget. At 5%, Frenet has the best mean change (-2.10%), while GFMI is close to Random (-5.09% vs. -4.79%). At 10%, Random is best mean change (-4.14%), and GFMI remains intermediate (-6.95%). The GFMI advantage appears at 15%, where it improves over Random (-2.37% vs. -5.90%) and avoids the large degradation seen under Frenet (-19.62%) and NRS (-15.05%). At 20%, GFMI remains the best measurement-guided rule (-6.14%), while Frenet and NRS remain much lower (-21.33% and -15.50%). This budget-dependent pattern supports the main Section 6 claim that GFMI becomes most useful as the pruning budget grows.

Table D.6 reports the cell values underlying Figure D.5. These values show that the high-budget GFMI advantage is not solely a consequence of averaging. At 15% and 20%, many GFMI cells remain close to zero, while the other measurement-guided rules include larger negative entries. The same pattern appears in the medians in Table 4: GFMI is closest to zero at every budget, including -1.31% at 15% and -3.27% at 20%, compared with Random's -5.41% at both budgets. These cell-level results show that GFMI is most effective when the budget is large enough for random or local-change-based rules to remove high-impact blocks, while final-layer graph alignment still identifies layers whose removal is less damaging.

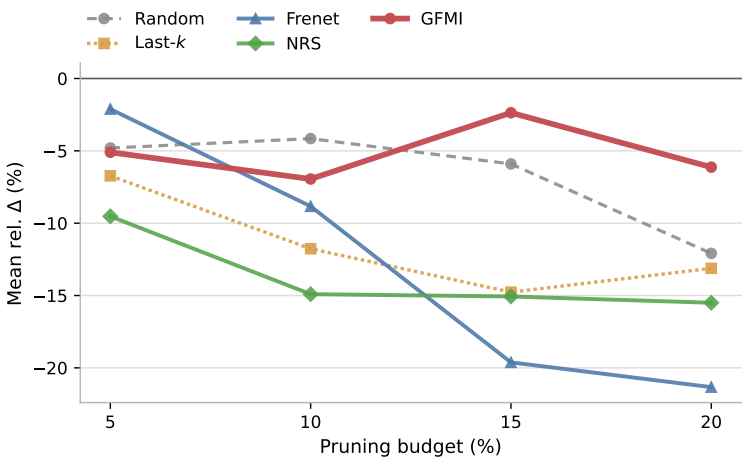


Figure D.5: Mean pruning curves over the 15 model-task cells. Values closer to zero are better.

Table D.6: Per-cell pruning relative score change. Each entry is the percentage change from the unpruned score for one model, task, rule, and budget.

Model / task	Rule	5%	10%	15%	20%
BGE-base					
AmazonPolarity	Random	-2.0	-2.0	-5.4	-5.4
	Last-k	-2.0	-2.0	-8.7	-8.7
	Frenet	-2.0	-2.0	-3.0	-3.0
	NRS	-0.6	-0.6	-5.4	-5.4
	GFMI	-2.0	-2.0	-3.3	-3.3
FiQA2018	Random	-18.6	-18.6	-26.9	-26.9
	Last-k	-18.6	-18.6	-37.8	-37.8
	Frenet	-18.6	-18.6	-36.9	-36.9
	NRS	-11.1	-11.1	-19.3	-19.3
	GFMI	-8.5	-8.5	-11.6	-11.6
STSBenchmark	Random	-2.9	-2.9	-6.1	-6.1
	Last-k	-2.9	-2.9	-6.8	-6.8
	Frenet	-0.8	-0.8	-5.5	-5.5
	NRS	-0.7	-0.7	-1.6	-1.6
	GFMI	-0.2	-0.2	-1.3	-1.3
E5-large					
AmazonPolarity	Random	+1.3	-0.1	-0.5	-1.7
	Last-k	+1.0	-5.5	-20.4	-26.8
	Frenet	+0.7	-5.3	-15.6	-14.2
	NRS	+1.0	-5.5	-7.3	-14.5
	GFMI	+0.3	+0.0	-0.8	-2.2
FiQA2018	Random	-0.3	-0.6	-0.4	-3.7
	Last-k	-31.8	-73.4	-87.4	-96.6
	Frenet	-10.0	-20.4	-80.1	-83.1
	NRS	-31.8	-73.4	-75.4	-72.9
	GFMI	-0.8	-2.0	-7.2	-8.9
STSBenchmark	Random	-0.1	-0.5	-0.7	-1.6
	Last-k	-1.4	-5.5	-19.3	-26.8
	Frenet	-0.1	-9.9	-36.2	-30.8
	NRS	-1.0	-5.5	-19.3	-26.8

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Model / task	Rule	5%	10%	15%	20%
	GFMI	-0.0	-0.1	-0.4	-0.8
LLM2Vec					
AmazonPolarity	Random	-1.6	-3.3	-4.5	-4.7
	Last-k	+1.8	+1.5	+1.6	+2.1
	Frenet	+1.0	+1.2	-2.4	-5.0
	NRS	-1.0	-0.7	-1.0	-0.8
	GFMI	+1.1	+1.5	+1.6	+2.1
FiQA2018	Random	+2.4	-4.4	-8.8	-3.8
	Last-k	+6.9	+30.1	+40.7	+34.1
	Frenet	-10.4	-12.5	-18.8	-26.5
	NRS	-7.6	-11.4	-31.8	-38.4
	GFMI	+6.9	+30.1	+40.7	+34.1
STSBenchmark	Random	+1.5	+0.2	-3.6	+0.9
	Last-k	+5.3	+9.2	+10.1	+9.9
	Frenet	-1.0	-2.3	-5.2	-6.7
	NRS	+5.3	+9.2	+11.2	+6.3
	GFMI	+3.1	+9.2	+10.1	+7.1
Nemotron					
AmazonPolarity	Random	-4.7	-7.9	-7.9	-17.3
	Last-k	-1.9	-9.2	-9.2	-5.9
	Frenet	-6.1	-9.0	-9.0	-8.7
	NRS	-1.9	-9.2	-9.2	-5.9
	GFMI	-6.1	-10.9	-10.9	-15.5
FiQA2018	Random	-6.6	-40.4	-40.4	-71.5
	Last-k	-18.4	-35.4	-35.4	-43.8
	Frenet	-13.6	-19.5	-19.5	-37.5
	NRS	-18.4	-35.4	-35.4	-43.8
	GFMI	-31.4	-52.8	-52.8	-54.5
STSBenchmark	Random	-2.4	-8.4	-8.4	-12.2
	Last-k	-3.4	-6.6	-6.6	-9.9
	Frenet	-0.1	-1.9	-1.9	-7.2
	NRS	-3.4	-6.6	-6.6	-9.9
	GFMI	-3.4	-6.6	-6.6	-5.2
Mistral-7B					
AmazonPolarity	Random	+4.4	+3.6	+3.7	+4.1
	Last-k	+4.5	+4.6	+4.7	+5.1
	Frenet	+4.6	+4.1	+5.3	+3.4
	NRS	+4.4	+3.9	+3.3	+4.0
	GFMI	+4.5	+0.9	+4.7	+5.1
FiQA2018	Random	-40.5	+64.9	+30.5	-16.8
	Last-k	-42.7	-83.2	-39.7	-29.0
	Frenet	+28.2	+2.3	-50.4	-37.4
	NRS	-42.7	-83.2	-39.7	-29.0
	GFMI	-42.7	-83.2	-39.7	-29.0
STSBenchmark	Random	-2.0	-41.9	-9.2	-14.6
	Last-k	+2.7	+20.4	-7.3	+44.1
	Frenet	-3.5	-37.9	-15.1	-20.9
	NRS	-33.4	+6.6	+11.5	+25.5
	GFMI	+2.7	+20.4	+42.0	-8.0